

PEDIATRIC CARDIOVASCULAR DISEASE — Revision Table

Bold = high-yield. [bracket] = standard added (slide image-only).

OUTFLOW OBSTRUCTION (well child)	
Lesion	Key points
Aortic stenosis	Valvular/subvalvular/supravalvular (5%). Severe → fatigue, exertional chest pain, syncope. Critical infant AS = duct-dependent (PDA keeps coronary/aortic flow)
Pulmonary stenosis	RV outflow obstruction; severe → RV failure. [ESM left upper sternal edge; balloon valvuloplasty]
Coarctation of aorta	[Radio-femoral delay, weak femoral pulse, UL hypertension, rib notching] ; neonate duct-dependent → prostaglandin]
PDA	[Continuous machinery murmur , bounding pulses; indomethacin/ibuprofen or device closure]

HEART FAILURE	
Aspect	Key points
Definition	Heart can't meet metabolic CO needs. CO = HR × SV (preload, afterload, contractility)
Causes	Rhythm (HB, SVT, VT); volume overload (VSD, PDA, regurgitation) , anaemia, sepsis; pressure overload (AS, PS, coarctation, HTN); systolic (myocarditis, DCM); diastolic (HCM, RCM, tamponade)
Clinical infant	Feeding difficulty + sweating, poor weight gain, irritability, respiratory distress
Clinical child	Fatigue, effort intolerance, anorexia/abdo pain, dyspnoea, orthopnoea
Signs	↑JVP, hepatomegaly , oedema, basal creps, cardiomegaly, gallop , holosystolic murmur
Ix	CXR (cardiomegaly), ECG, echo (function) , blood gas (acidosis)
Treatment	Treat cause; semi-upright/infant chair; ↑calorie + no added salt ; diuretics, inotropes, afterload reducers

KAWASAKI DISEASE (children <5yr)	
Aspect	Key points
Diagnosis	Fever ≥5 days + 4 of 5 : 1. Non-purulent red eyes 2. Lips/strawberry tongue 3. Hands/feet red→ peeling 4. Body rash (groin) 5. Cervical node (unilateral >1.5cm)
Treatment	Aspirin + IVIG ± steroids
Complication	Coronary artery changes + aneurysms
Prognosis	95% complete recovery if no coronary damage on echo

ARRHYTHMIAS	
Arrhythmia	Key points
SVT	Commonest childhood arrhythmia ; HR 250–300 ; re-entry; HF in infant, hydrops fetalis. WPW = short PR + delta wave . Mgmt: vagal (ice/face 80%) → IV adenosine (TOC) → DC cardioversion (0.5–2 J/kg). Maintenance flecainide/sotalolol; ablation if WPW relapse
Long QT	Syncope on exertion/stress (mistaken for epilepsy); AD channelopathy (Na/K/Ca); sudden death from VT; assoc erythromycin/electrolytes

RHEUMATIC FEVER & INFECTIVE ENDOCARDITIS	
Topic	Key points
Rheumatic fever	Autoimmune to group A strep ; age 5–15; → RHD; overcrowding/low-income
Infective endocarditis	All CHD risk except secundum ASD ; highest = turbulent jet (VSD, coarctation, PDA) / prosthetic . Signs: fever, splinter haemorrhages, splenomegaly, microscopic haematuria , clubbing (late). Blood cultures BEFORE antibiotics ; echo vegetations; Strep viridans commonest; penicillin + aminoglycoside 6 wks IV